Examiner only

**6.** Polymer gels are commonly used in disposable nappies.

A company that manufactures disposable nappies was investigating the effect of temperature on the mass of water the polymer gel in their nappies is able to absorb.

(a) The results collected using water at 40 °C are given below. The initial mass of the polymer gel bead was 0.035 g.

| Time (hours) | Mass of bead (g) | Mass of water absorbed by bead (g) (to 1 decimal place) |
|---|---|---|
| 0 | 0.035 | 0.0 |
| 2 | 4.048 | 4.0 |
| 4 | 6.030 | 6.0 |
| 6 | 7.280 | 7.2 |
| 8 | 7.891 | 7.9 |
| 10 | 8.181 | 8.1 |
| 12 | 8.181 | 8.1 |

(i) The percentage increase in the mass of the bead is calculated using the following equation.

$$\text{percentage increase} = \frac{\text{mass of water absorbed}}{\text{initial mass of bead}} \times 100$$

Calculate the percentage increase in the mass of the bead after 2 hours.
Give your answer to the nearest whole number. [1]

Percentage increase = .............................................. %

(ii) What property of polymer gels does the figure calculated in part (i) demonstrate? [1]

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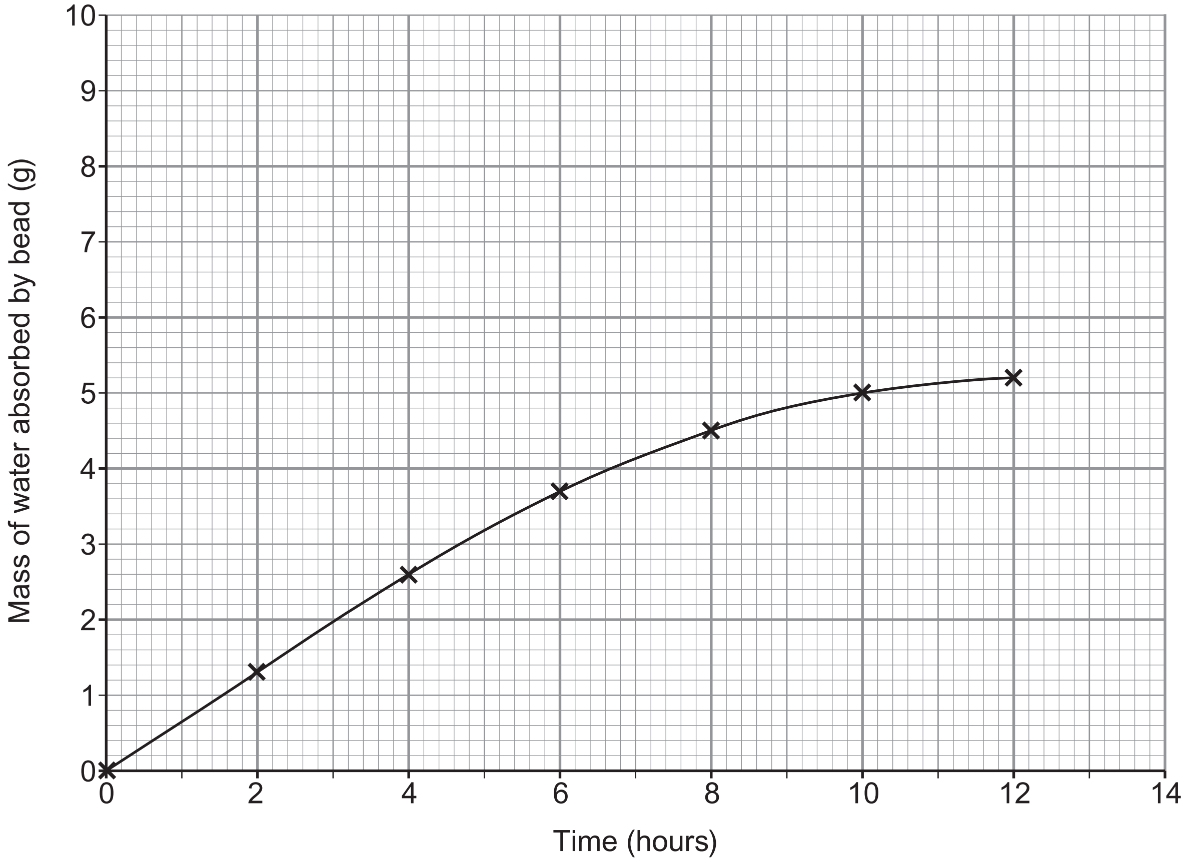
(b) (i) On the grid below, plot the results using water at 40 °C and draw a suitable line. Use the mass of water absorbed by the bead to 1 decimal place.

The results using water at 10 °C have already been plotted. [3]




(ii) Give **two** differences between the absorbing properties of the bead using water at 10 °C and at 40 °C. [2]

Difference 1 ....................................................................................................

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Difference 2 ....................................................................................................

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